

High Low Game

We want you to gain more experience working with control flow and Booleans in Python. To do this, we are going to have you develop a game! The game is called *High-Low* and the way it's played goes as follows:

1. Two numbers are generated from 1 to 100 (inclusive on both ends): one for you and one for a computer, who will be your opponent. You get to see your number, but not the computer's!
2. You make a guess, saying your number is either higher than or lower than the computer's number
3. If your guess matches the truth (ex. you guess your number is higher, and then your number is actually higher than the computer's), you get a point!

These steps make up one *round* of the game. The game is over after all rounds have been played.

Here's a sample run of the game where we play three rounds (user input is in blue for visual clarity):

```
Welcome to the High-Low Game! -----
Round 1
Your number is 8
Do you think your number is higher or lower than the computer's?: lower
You were right! The computer's number was 35
Your score is now 1
Round 2
Your number is 88
Do you think your number is higher or lower than the computer's?: higher
Aww, that's incorrect. The computer's number was 100
Your score is now 1
Round 3
Your number is 63
Do you think your number is higher or lower than the computer's?: higher
You were right! The computer's number was 5
Your score is now 2
Thanks for playing!
```

To help out, we've made some milestones to guide you.

Milestone #1: Generate the random numbers

Generate the random numbers for you and the computer. For now, print both of them out to help you test the logic in later milestones. Your output should look like this (with different random numbers):

```
Welcome to the High-Low Game! -----  
The computer's number is 23  
Your number is 82
```

Milestone #2: Get user choice

Get user input for their choice of whether they think their number is higher or lower than the computer's number.

```
Welcome to the High-Low Game! -----  
The computer's number is 7  
Your number is 17  
Do you think your number is higher or lower than the computer's?: higher
```

Milestone #3: Write the game logic

Write code that maps out all the ways to win the round and prints out the results. When does the user win? How might we check for this? (Note: If you and the computer share the same number, the computer should win since your number wouldn't be higher nor lower.)

```
Welcome to the High-Low Game! -----  
The computer's number is 79 Y  
our number is 3  
Do you think your number is higher or lower than the computer's?: lower  
You were right! The computer's number was 79
```

(Example case for when you and the computer have the same number)

```
Welcome to the High-Low Game! -----  
The computer's number is 57  
Your number is 57  
Do you think your number is higher or lower than the computer's?: lower Aww,  
that's incorrect.  
The computer's number was 57
```

Milestone #4: Play multiple rounds

Milestones 1-3 make up a single round of the game. Now that your game logic is sound, you can remove the line printing out the computer's number. Next, write code to play multiple rounds of the game! How many rounds should they play you ask? We've provided you with the `NUM_ROUNDS` constant. `NUM_ROUNDS` is the number of rounds you should have the user play. For reference, in the first example, we had `NUM_ROUNDS = 3`. After each round, add a blank line to separate the rounds visually. Make sure to print out the round number as well!

```
Welcome to the High-Low Game! -----
Round 1 Your number is 22
Do you think your number is higher or lower than the computer's?: lower
You were right! The computer's number was 23
Round 2
Your number is 76
Do you think your number is higher or lower than the computer's?: higher
Aww, that's incorrect. The computer's number was 81 ...
(more rounds are played)
```

Milestone #5: Adding a points system and a thank you!

Keep track of the player's score! You should print out the player's score after each round. After they finish all the rounds, say thank you! After this step, you will have coded up the entire game!

```
Welcome to the High-Low Game! -----
Round 1 Your number is 22
Do you think your number is higher or lower than the computer's?: lower
You were right! The computer's number was 23
Your score is now 1

Round 2 Your number is 76
Do you think your number is higher or lower than the computer's?: higher
Aww, that's incorrect. The computer's number was 81
Your score is now 1
... (more rounds are played)
Round 5
Your number is 88
Do you think your number is higher or lower than the computer's?: higher
You were right! The computer's number was 35
Your score is now 3

Thanks for playing!
```

Extension #1: Safeguard user input

Add safety precautions that make sure the player can only input 'higher' or 'lower' as an answer. If they enter anything else, prompt them again and let them know there's only two options!

Welcome to the High-Low Game! -----

Round 1 Your number is 1

Do you think your number is higher or lower than the computer's?: *even*

Please enter either higher or lower: *lower*

You were right! The computer's number was 6

Your score is now 1

Extension #2: Conditional ending messages

Add conditional messages at the end which gauge players on how they performed! For ours, we evaluated the player and gave them separate messages if:

- they had a perfect game
- they won at least half the rounds (rounded down)
- they won less than half the rounds

For example, if the user played five rounds, here's the messages we chose:

(Won 5 rounds)

... Your score is now 5

Wow! You played perfectly!

(Won at least half the rounds, rounded down)

... Your score is now 2

Good job, you played really well!

(Won less than half the rounds)

... Your score is now 1

Better luck next time!